Gloss → Healthcare Text — The Language Layer

Per-model deep dive · where BLEU, ROUGE, METEOR and WER actually apply

Task: turn a recognized gloss (e.g. 'pain where') into a natural sentence ('Where is the pain?')
Models: exact rule-based dictionary · fuzzy token-overlap matcher · mT5 (planned)

1 Why this layer needs different metrics

Recognition outputs a *label*; this layer outputs a *sentence*. That is why the translation metrics your review asked for — BLEU, ROUGE, METEOR — live here and not on the recognition models. Each compares generated text to a reference: BLEU on n-gram precision, ROUGE on n-gram/subsequence recall, METEOR on unigram matches with a word-order penalty, and WER on word-level edits. Reporting them on a classifier's single-label output would be a category error.

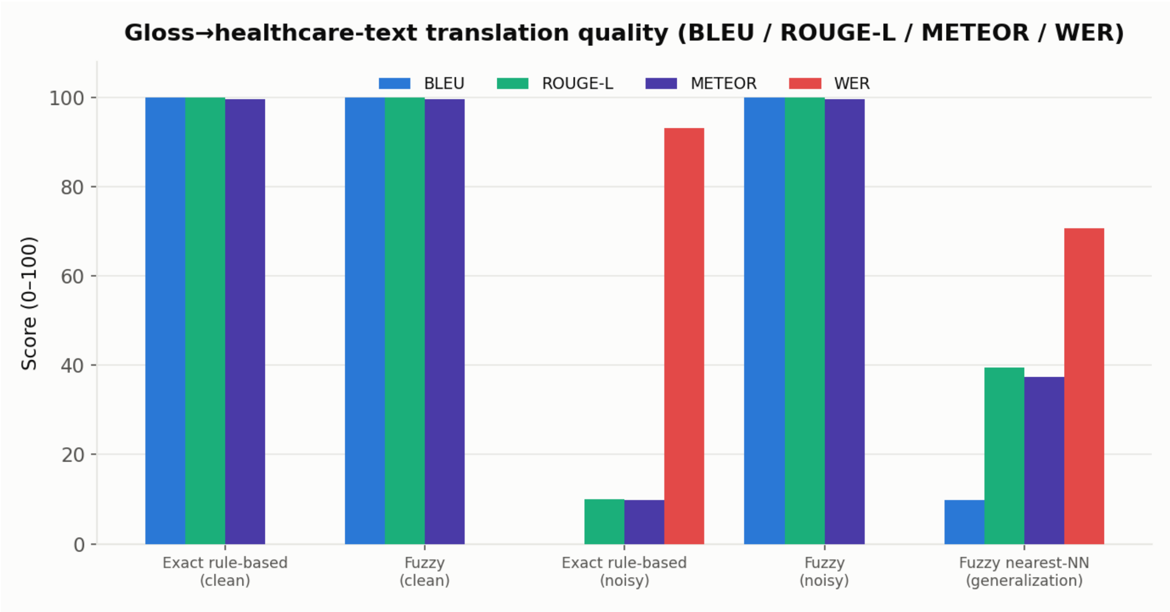
2 Method of execution

Two baseline translators are evaluated on the 20-phrase healthcare set. The **exact rule-based** translator looks up the normalized gloss in a dictionary. The **fuzzy token-overlap** translator scores each stored gloss by Jaccard token overlap and returns the best match above a threshold — so it tolerates reordered or partially-dropped words. Both are tested three ways: on clean gloss input, on simulated noisy recognizer output (words reordered / dropped), and on a leave-one-out generalization test where the exact phrase is removed and the translator must retrieve the nearest *different* one.

3 Results

Condition / translator	BLEU	R-1	R-2	R-L	METEOR	WER↓	Exact
Exact rule-based — clean	100.0	100.0	100.0	100.0	99.5	0.0	100%
Fuzzy overlap — clean	100.0	100.0	100.0	100.0	99.5	0.0	100%
Exact rule-based — noisy	0.0	10.0	10.0	10.0	9.9	93.1	10%
Fuzzy overlap — noisy	100.0	100.0	100.0	100.0	99.5	0.0	100%
Fuzzy nearest-NN — generalization	**9.8**	**39.5**	**27.3**	**39.5**	**37.3**	**70.6**	**0%**

All scores 0–100. R-1/R-2/R-L = ROUGE-1/2/L. Highlighted row = the generalization test.



Translation quality across the three conditions.

4 Why these results

On known phrases both translators are perfect (BLEU 100, WER 0). The moment input is perturbed — which is what an imperfect recognizer produces — the exact matcher collapses (BLEU 0, WER 93, 10% coverage) while the fuzzy matcher recovers every phrase (BLEU 100, WER 0). **This is the empirical case for order-insensitive matching downstream of recognition.** The generalization row is the honest one: asked for a phrase it has never seen, the lookup translator retrieves a near-neighbour, scoring BLEU 9.8 / ROUGE-L 39.5 / METEOR 37.3 — the metrics correctly registering a partial match. A dictionary cannot compose novel wordings; that is the precise gap a trainable model fills.

> **What makes it stronger / weaker, and the next step**
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> **Stronger:** the fuzzy matcher is simple, fast, and robust to recognizer noise — the right choice for an MVP on a fixed phrase set. **Weaker:** both translators are closed-set lookups; they cannot generalize to unseen phrasings, so BLEU/ROUGE/METEOR on novel input are low. The planned fix — already scaffolded as T5-format training data — is to fine-tune **mT5-small** for English/Kiswahili/KSL gloss→text, which can generate rather than retrieve. These BLEU/ROUGE/METEOR numbers become the baseline that mT5 must beat.

5 Role in the hybrid pipeline

This layer sits between recognition and the doctor: recognized gloss → healthcare sentence. In the recommended pipeline it runs the fuzzy matcher today (robust to the recognizer's mistakes) and a fine-tuned mT5 next. Note the two stages are evaluated separately by necessity — recognition is benchmarked on WLASL words, the language layer on healthcare phrases — so the honest end-to-end estimate is the product of the two stages' quality, not a single trained number. Closing that gap requires a KSL healthcare video corpus, which is the project's central data need.